

COURSE NAME & CODE: ECA 0303 – Signals and systems

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Experiment 5.1

Impulse Response of the System

Aim:

determine and plot the impulse response of a continuous-time system. This involves applying an impulse function $\delta(t)$ as the input and observing the system's output.

Program:

```
% Test Case: Impulse Response of a Continuous-Time System
```

```
// Define time range and sampling interval
```

```
t_min = -1; // Start time
```

```
t_max = 5; // End time
```

```
dt = 0.01; // Time step
```

```
t = t_min:dt:t_max; // Time vector
```

```
// Define the impulse function
```

```
x_impulse = zeros(1, length(t));
```

```
x_impulse(t == 0) = 1; // Approximate impulse function
```

```
// Initialize the output signal with zeros
```

```
y_impulse = zeros(1, length(t));
```

```
// Define the system parameters
```

```
a = 2; // Coefficient for y(t)
```

```
b = 1; // Coefficient for x(t)
```

```
// Compute the impulse response using the Laplace Transform approach
```

```
// Analytical solution: y(t) = (b/a) * exp(-a*t) for t >= 0
```

```

for i = 1:length(t)
    if t(i) >= 0
        y_impulse(i) = (b / a) * exp(-a * t(i));
    end
end

// Plot the input and output signals for impulse response
clf; // Clear the current figure
subplot(2,1,1);
plot(t, x_impulse, 'b');
xlabel('t (Time)');
ylabel('x(t)');
title('Input Signal x(t) - Impulse Function');
subplot(2,1,2);

plot(t, y_impulse, 'r');
xlabel('t (Time)');
ylabel('y(t)');
title('Impulse Response y(t) of the System');

```

Output:



Result:

• 1. Impulse Function:

o The plot should show a sharp peak at $t=0$ for the impulse function, indicating the instantaneous nature of the input.

2. Impulse Response:

o the plot should display the system's response to the impulse input, which typically starts at 0 and exponentially decays over time. This provides insight into how the system responds to a brief, instantaneous input.

This code will help you visualize the impulse response of the continuous time system and understand its behavior..